Module III — Semiconductor properties

Derivations, step by step

Conductivity of a semiconductor: $\sigma = e(n\mu_n + p\mu_p)$

4 marks

v_d: drift velocity. μ: mobility, the drift velocity per unit field. n, p: electron and hole concentrations; μ_n, μ_p their mobilities. $\mathcal{E}$: applied field.

BEFORE YOU START: WHAT THIS USES

v_d	drift velocity, the average velocity gained from the field: Drude derivation
μ	mobility, drift velocity per unit field: drift velocity and mobility
n, p	electrons in the conduction band and holes in the valence band: intrinsic semiconductors
$J = \sigma\mathcal{E}$	Ohm's law in local form: the conductivity σ is defined by it
e	magnitude of the electron charge; holes carry $+e$: holes

START

$$v_d = \mu\mathcal{E}$$

RESULT

$$\sigma_i = en_i(\mu_n + \mu_p)$$

DRIFT

$$v_d = \mu\mathcal{E}$$

Carriers are accelerated by the field but keep colliding with the lattice (phonons, impurities). On average they gain only a small extra velocity, proportional to the field: $v_d = e\tau\mathcal{E}/m$ with the relaxation time τ, as in the Drude derivation. The constant $e\tau/m$ is the mobility μ.

ELECTRON CURRENT

$$\text{charge through area } A \text{ in } dt = n\,e\,(A\,v_{dn}\,dt)$$

Every carrier within $v_{dn}\,dt$ of the area gets through in time dt

They fill a box of base A and length $v_{dn}\,dt$, which holds $nAv_{dn}\,dt$ carriers of charge e each.

$$J_n = nev_{dn}$$

Current per unit area: divide the charge by $A\,dt$

n electrons per unit volume, each carrying charge e at the drift velocity.

$$J_n = ne\mu_n\mathcal{E}$$

substitute: $v_{dn} = \mu_n\mathcal{E}$

$$\sigma_n = ne\mu_n$$

Compare with Ohm's law $J = \sigma\mathcal{E}$

HOLES ADD TO IT

$$\sigma = \sigma_n + \sigma_p = e(n\mu_n + p\mu_p)$$

add: $\sigma_p = pe\mu_p$

Holes (what a hole is) drift the other way, but their charge is $+e$, so their current is in the same direction as the electrons' current. The two conductivities add.

Check. Electrons and holes drift in opposite directions. Do their currents add or cancel?

- ✓ **add: opposite charge and opposite motion give current in the same direction**
- cancel exactly
- only the faster one counts
- they cancel in an intrinsic sample

Current is charge × velocity; both signs flip for holes, so the product has the same sign.

INTRINSIC SAMPLE

$$\sigma_i = en_i(\mu_n + \mu_p)$$

substitute: $n = p = n_i$

$\mu_n > \mu_p$ because a hole moves by electrons hopping between bonds and has the larger effective mass. Si at 300 K: $\mu_n = 0.135$, $\mu_p = 0.048\ \mathrm{m^2V^{-1}s^{-1}}$, $n_i = 1.5 \times 10^{16}\ \mathrm{m^{-3}}$ give $\sigma_i = 4.4 \times 10^{-4}\ \mathrm{S\,m^{-1}}$.

$\ln\sigma$ against $1/T$ measures the band gap

4 marks

Mobilities taken as independent of temperature. n_i from the law of mass action.

BEFORE YOU START: WHAT THIS USES

$\sigma_i = en_i(\mu_n + \mu_p)$	intrinsic conductivity: conductivity derivation
$n_i = (N_cN_v)^{1/2}e^{-E_g/2kT}$	from the law of mass action
$\mu_n,\ \mu_p$	mobilities, taken as independent of temperature here
$N_cN_v \propto T^3$	shown in the step below

START

$$\sigma = en_i(\mu_n + \mu_p)$$

RESULT

$$\ln\sigma = -\frac{E_g}{2k}\cdot\frac{1}{T} + \frac{3}{2}\ln T + \text{const}$$

START

$$\sigma = en_i(\mu_n + \mu_p)$$

The intrinsic conductivity.

$$\sigma = e(\mu_n + \mu_p)(N_cN_v)^{1/2}e^{-E_g/2kT}$$

substitute: n_i

$$(N_cN_v)^{1/2} = 2\left(\frac{2\pi k}{h^2}\right)^{3/2}(m_n^*m_p^*)^{3/4}\,T^{3/2}$$

Put in N_c *and* N_v *(electron, hole derivations): each carries* $T^{3/2}$

Only T changes with temperature; the masses are constants.

$$\sigma = CT^{3/2}e^{-E_g/2kT}$$

$N_cN_v \propto T^3$*: collect the constants into* C

TAKE LOGS

$$\ln \sigma = \ln C + \frac{3}{2}\ln T - \frac{E_g}{2kT}$$

ln

$$\ln \sigma = -\frac{E_g}{2k}\cdot\frac{1}{T} + \frac{3}{2}\ln T + \text{const}$$

rearrange

The $1/T$ term dominates: $\ln\sigma$ against $1/T$ is nearly a straight line of slope $-E_g/2k$. This is how a band gap is measured.

Check. The slope of $\ln\sigma$ against $1/T$ is -5800 K. What is E_g?

- ✓ $2k \times 5800\ \text{K} \approx 1.0$ eV
- $k \times 5800\ \text{K} \approx 0.5$ eV
- $5800\ \text{K}/2k$
- 0.26 eV

Slope $= -E_g/2k$, so $E_g = 2k \times 5800 = 2 \times 8.62 \times 10^{-5} \times 5800 \approx 1.0$ eV.

Electron concentration in the conduction band

8 marks

$D(E)$: density of states per unit volume. $f(E)$: Fermi function. m_n^*: effective mass of electrons in the conduction band. E_c: bottom of the conduction band. Kinetic energy of an electron at E is $E - E_c$.

BEFORE YOU START: WHAT THIS USES

$D(E)$	states per unit energy and volume, from the density-of-states derivation (the version with h is its last step)
$f(E)$	Fermi function, probability that a state is occupied: the Fermi function
$E_c,\ E_v,\ E_g$	band edges and gap: metals, semiconductors, insulators
m_n^*	effective mass of an electron in the conduction band: effective-mass derivation
E_F	Fermi level, lying inside the gap: the recipe: states times occupation
$\int_0^\infty x^{1/2}e^{-x}dx$	the integral is evaluated in the steps below

START

$$dn = D(E)f(E)\,dE$$

RESULT

$$n = N_c\, e^{-(E_c - E_F)/kT}$$

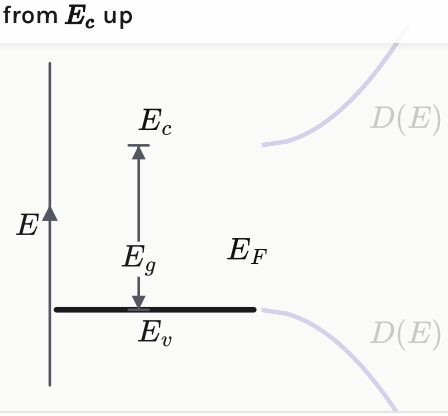
$$N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$$

COUNT OCCUPIED STATES

$$dn = D(E)\,f(E)\,dE$$

Electrons between E and $E + dE$ = (states there) × (chance each is occupied).

Electrons fill the conduction band from E_c up

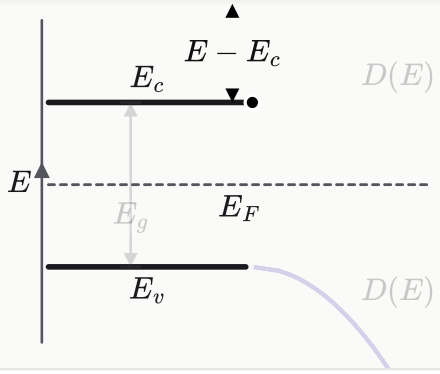


$$D(E) = \frac{4\pi}{h^3}(2m_n^*)^{3/2}(E - E_c)^{1/2}$$

Free-electron $D(E)$, with $m \to m_n^*$ and $E \to E - E_c$

The free-electron result $\frac{4\pi}{h^3}(2m)^{3/2}E^{1/2}$ per unit volume (derived in Module II) counts kinetic energy from zero. In the conduction band the electron behaves as a free particle of mass m_n^* (effective mass) with kinetic energy counted from E_c.

$D(E) \propto \sqrt{E - E_c}$



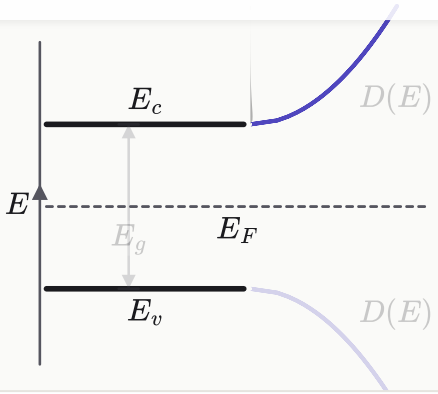
BOLTZMANN TAIL

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \approx e^{-(E-E_F)/kT}$$

For $E - E_F \gg kT$, drop the 1

E_F lies in the gap, several kT below E_c ($kT = 0.026$ eV at 300 K), so the exponential is huge for every conduction-band state and the 1 is negligible. This is the classical Boltzmann form of the Fermi function.

Occupied states: $D(E) \times f(E)$



Check. Why can we drop the $+1$ in the Fermi function here?

- ✓ $E - E_F \gg kT$ for every state in the conduction band
- E_F is above E_c
- kT is large
- the band is full

With $E - E_F$ several kT, $e^{(E-E_F)/kT}$ is very much greater than 1.

INTEGRATE OVER THE BAND

$$n = A \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-(E-E_F)/kT}\, dE$$

Add up from E_c to ∞, writing $A = \frac{4\pi}{h^3}(2m_n^*)^{3/2}$

$$n = A e^{-(E_c-E_F)/kT} \int_{E_c}^{\infty} (E - E_c)^{1/2} e^{-(E-E_c)/kT}\, dE$$

Split $E - E_F = (E - E_c) + (E_c - E_F)$ and take the constant out

SUBSTITUTE $x = (E - E_c)/kT$

$$n = A(kT)^{3/2} e^{-(E_c-E_F)/kT} \int_0^{\infty} x^{1/2} e^{-x}\, dx$$

substitute: $E - E_c = kTx,\ dE = kT\,dx$

$(E - E_c)^{1/2} dE = (kT)^{3/2} x^{1/2} dx$; the limits become 0 and ∞.

THE INTEGRAL OVER x

$$\int_0^{\infty} x^{1/2} e^{-x} dx = \frac{1}{2} \int_0^{\infty} x^{-1/2} e^{-x} dx$$

By parts: $u = x^{1/2},\ dv = e^{-x}dx$

The boundary term $\left[-x^{1/2}e^{-x}\right]_0^{\infty}$ is zero and what is left is $\int_0^{\infty} \frac{1}{2} x^{-1/2} e^{-x} dx$.

$$\int_0^\infty x^{-1/2}e^{-x}\,dx = 2\int_0^\infty e^{-y^2}\,dy$$

substitute: $x = y^2,\ dx = 2y\,dy$

$$\int_0^\infty e^{-y^2}\,dy = \frac{\sqrt{\pi}}{2}$$

The Gaussian integral

Call it I and square it: $I^2 = \int_0^\infty\int_0^\infty e^{-(x^2+y^2)}dx\,dy$. In polar coordinates over the quarter plane, $I^2 = \int_0^{\pi/2} d\phi \int_0^\infty e^{-r^2} r\,dr = \frac{\pi}{2}\cdot\frac{1}{2} = \frac{\pi}{4}$.

$$\int_0^\infty x^{1/2}e^{-x}\,dx = \frac{1}{2}\cdot 2\cdot\frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{2}$$

combine

$$n = A(kT)^{3/2}e^{-(E_c-E_F)/kT}\cdot\frac{\sqrt{\pi}}{2}$$

evaluate: $\int_0^\infty x^{1/2}e^{-x}dx = \sqrt{\pi}/2$

Put the value just derived into the integral.

TIDY

$$n = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} e^{-(E_c-E_F)/kT}$$

simplify

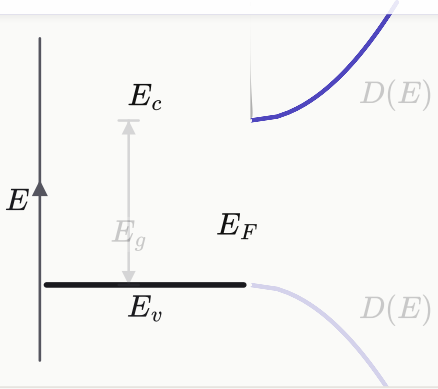
$A(kT)^{3/2}\frac{\sqrt{\pi}}{2} = \frac{4\pi}{h^3}\cdot\frac{\sqrt{\pi}}{2}(2m_n^*kT)^{3/2} = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$.

$$n = N_c\, e^{-(E_c-E_F)/kT}$$

define: $N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$

N_c is the effective density of states at the band edge: it is as if all the states were squeezed onto E_c, each occupied with probability $e^{-(E_c-E_F)/kT}$. For Si, $N_c \approx 2.8\times10^{25}(T/300)^{3/2}\ \mathrm{m}^{-3}$. The result holds for doped samples too.

$n = N_c e^{-(E_c-E_F)/kT}$



Hole concentration in the valence band

5 marks

$1-f(E)$: probability that a state is empty, i.e. holds a hole. m_p^*: hole effective mass. E_v: top of the valence band; a hole at E has kinetic energy $E_v - E$.

BEFORE YOU START: WHAT THIS USES

$1 - f(E)$	probability that a state is empty, that is, holds a hole: holes
m_p^*	hole effective mass: holes
E_v	top of the valence band; a hole at E has kinetic energy $E_v - E$
$D(E)$, $\sqrt{\pi}/2$	same density of states and same integral as in the electron derivation

START

$$dp = D(E)\,[1 - f(E)]dE$$

RESULT

$$p = N_v\, e^{-(E_F - E_v)/kT}$$

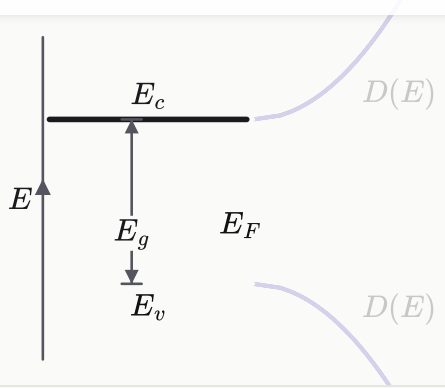
$$N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

COUNT EMPTY STATES

$$dp = D(E)\,[1 - f(E)]dE$$

A hole is a state the electrons have left empty.

Holes: empty states below E_v



PROBABILITY OF AN EMPTY STATE

$$1 - f(E) = 1 - \frac{1}{e^{(E - E_F)/kT} + 1} = \frac{e^{(E - E_F)/kT}}{e^{(E - E_F)/kT} + 1}$$

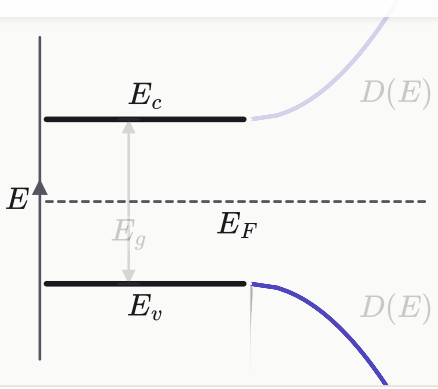
Put over a common denominator

$$1 - f(E) = \frac{1}{e^{(E_F - E)/kT} + 1} \approx e^{-(E_F - E)/kT}$$

For $E_F - E \gg kT$

Divide top and bottom by $e^{(E - E_F)/kT}$ to get $1/(e^{(E_F - E)/kT} + 1)$. Holes become exponentially rarer as you go deeper into the valence band.

Empty states: $D(E) \times [1 - f(E)]$



Check. In the valence band $E < E_F$. Which form does $1 - f(E)$ take?

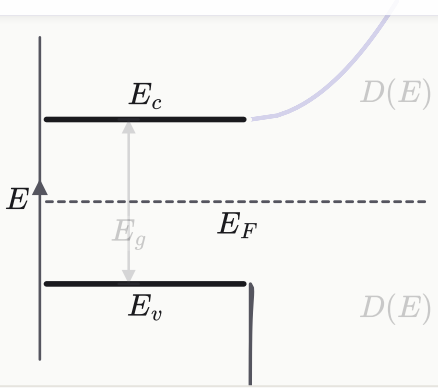
- ✓ $e^{-(E_F - E)/kT}$
- $e^{-(E - E_F)/kT}$
- 1
- $e^{-E/kT}$

$1 - \frac{1}{e^x + 1} = \frac{1}{e^{-x} + 1}$ with $x = (E - E_F)/kT < 0$; for $|x| \gg 1$ this is $e^x = e^{-(E_F - E)/kT}$.

$$D(E) = \frac{4\pi}{h^3}(2m_p^*)^{3/2}(E_v - E)^{1/2}$$

Same $D(E)$, with m_p^* and kinetic energy $E_v - E$

$D(E) \propto \sqrt{E_v - E}$



INTEGRATE DOWN THE BAND

$$p = A' \int_{-\infty}^{E_v} (E_v - E)^{1/2} e^{-(E_F - E)/kT}\, dE$$

Add up from $-\infty$ to E_v, with $A' = \frac{4\pi}{h^3}(2m_p^*)^{3/2}$

$$p = A'(kT)^{3/2} e^{-(E_F - E_v)/kT} \cdot \frac{\sqrt{\pi}}{2}$$

substitute: $E_v - E = kTx$

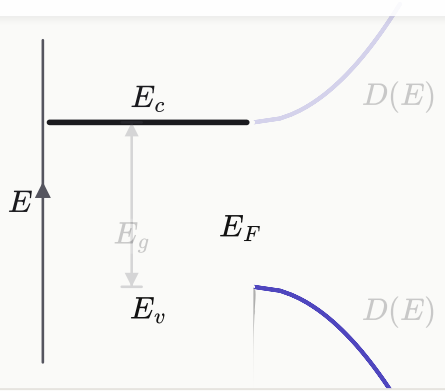
The same integral as for electrons, derived in the electron derivation.

$$p = N_v\, e^{-(E_F - E_v)/kT}$$

define: $N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$

The mirror image of the electron result, with m_p^* and the distance of E_F above E_v.

$p = N_v e^{-(E_F - E_v)/kT}$



Fermi level of an intrinsic semiconductor

5 marks

Intrinsic: $n = p = n_i$. N_c, N_v as in the two previous derivations.

BEFORE YOU START: WHAT THIS USES

$n = p = n_i$	in a pure sample every electron in the conduction band left a hole: intrinsic semiconductors
$N_c,\ N_v$	effective densities of states from the electron and the hole derivations
E_F	Fermi level: where it sits in a pure sample
$\ln x^a = a \ln x$	the logarithm rule used in the step with the ask

START

$$N_c e^{-(E_c - E_F)/kT} = N_v e^{-(E_F - E_v)/kT}$$

RESULT

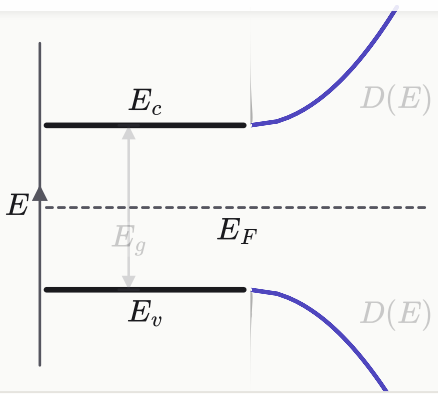
$$E_F = \frac{E_c + E_v}{2} + \frac{3kT}{4} \ln \frac{m_p^*}{m_n^*}$$

N = P

$$N_c e^{-(E_c - E_F)/kT} = N_v e^{-(E_F - E_v)/kT}$$

In a pure sample every conduction electron left a hole behind.

$n = p$



$$e^{(2E_F - E_c - E_v)/kT} = \frac{N_v}{N_c}$$

Collect the exponentials on one side

$$\frac{2E_F - E_c - E_v}{kT} = \ln \frac{N_v}{N_c}$$

ln

$$E_F = \frac{E_c + E_v}{2} + \frac{kT}{2} \ln \frac{N_v}{N_c}$$

rearrange

PUT IN N_c AND N_v

$$\frac{N_v}{N_c} = \left(\frac{m_p^*}{m_n^*} \right)^{3/2}$$

Everything in N_c, N_v cancels except the masses

Check. What is $\ln\left[(m_p^*/m_n^*)^{3/2}\right]$?

- ✓ $\frac{3}{2}\ln(m_p^*/m_n^*)$
- $\ln(m_p^*/m_n^*)^{2/3}$
- $(\ln m_p^* / \ln m_n^*)^{3/2}$
- $\frac{2}{3}\ln(m_p^*/m_n^*)$

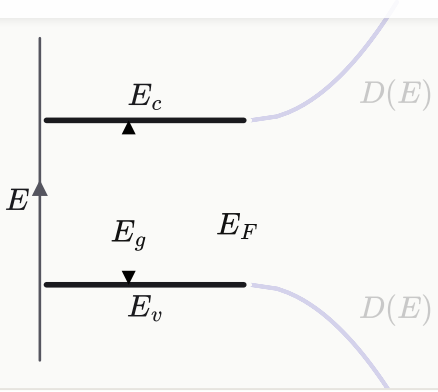
$\ln x^a = a \ln x$.

$$E_F = \frac{E_c + E_v}{2} + \frac{3kT}{4}\ln\frac{m_p^*}{m_n^*}$$

substitute

At 0 K, or whenever $m_p^* = m_n^*$, E_F is exactly at mid-gap. Usually $m_p^* > m_n^*$, so E_F rises slightly with T; for Si at 300 K by only about 0.01 eV.

E_F at the middle of the gap



Law of mass action and the intrinsic concentration n_i

5 marks

$E_g = E_c - E_v$: band gap. n, p from the two concentration derivations.

BEFORE YOU START: WHAT THIS USES

n, p	from the electron and the hole derivations
$E_g = E_c - E_v$	band gap: metals, semiconductors, insulators
n_i	intrinsic carrier concentration: intrinsic semiconductors
mass action	np fixed at a given temperature: the law and n_i

START

$$np = N_c N_v e^{-(E_c - E_F)/kT} e^{-(E_F - E_v)/kT}$$

RESULT

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2}(m_n^* m_p^*)^{3/4} e^{-E_g/2kT}$$

MULTIPLY N AND P

$$np = N_c N_v e^{-(E_c - E_F)/kT} e^{-(E_F - E_v)/kT}$$

Multiply the two results.

$$np = N_c N_v e^{-(E_c - E_v)/kT}$$

combine

E_F cancels.

Check. What happened to E_F?

- ✓ it cancelled: $-(E_c - E_F) - (E_F - E_v) = -(E_c - E_v)$
- it was set to zero
- it equals E_c
- it doubles

Add the two exponents: the E_F terms are $+E_F$ and $-E_F$.

$$np = N_c N_v e^{-E_g/kT} = n_i^2$$

substitute: $E_c - E_v = E_g$

Law of mass action: at a given temperature np is fixed, whether the sample is doped or not. Doping that raises n lowers p in proportion.

INTRINSIC CONCENTRATION

$$n_i = (N_c N_v)^{1/2} e^{-E_g/2kT}$$

root

Note the 2 in the exponent: $E_g/2kT$, not E_g/kT.

$$n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2} (m_n^* m_p^*)^{3/4} e^{-E_g/2kT}$$

Put in N_c *and* N_v

Pure Ge at 300 K: $n_i \approx 2.4 \times 10^{19}\ \mathrm{m}^{-3}$ out of 4.4×10^{28} atoms per m^3, about 5 broken bonds in 10^{10} atoms.

n-type: Fermi level and electron concentration

8 marks

N_d: donors per unit volume at level E_d, just below E_c. N_d^+: ionised donors. $\Delta E = E_c - E_d$: ionisation energy of a donor. Moderate temperature: holes are negligible, and E_F is a few kT from both E_c and E_d.

BEFORE YOU START: WHAT THIS USES

$N_d,\ E_d$	donor concentration and donor level just below E_c: extrinsic semiconductors
N_d^+	ionised donors: donors that have lost their electron
$n = N_c e^{-(E_c - E_F)/kT}$	electron concentration: the electron derivation
$p = n_i^2/n$	law of mass action: mass-action derivation
$1 - f(E_d)$	chance that the donor level is empty: how $1 - f$ becomes a Boltzmann factor
$\Delta E = E_c - E_d$	ionisation energy of a donor

START

$$n = N_d^+ + p \approx N_d^+$$

RESULT

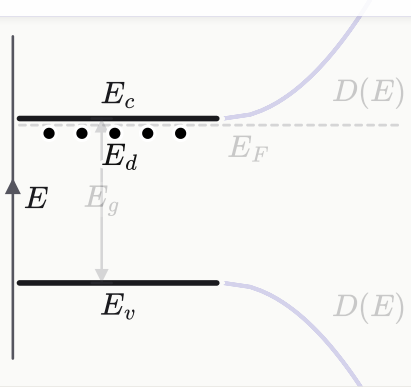
$$n = (N_d N_c)^{1/2} e^{-\Delta E/2kT}$$

NEUTRALITY

$$n = N_d^+ + p \approx N_d^+$$

Every conduction electron came from an ionised donor or from a broken bond. With enough donors, $p = n_i^2/n$ is tiny (mass action), so drop it. This is charge neutrality (extrinsic semiconductors).

Donor levels E_d just below E_c



IONISED DONORS

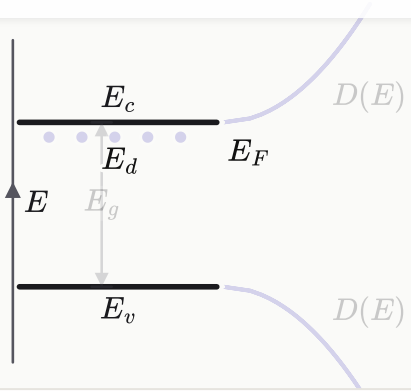
$$N_d^+ = N_d\,[1 - f(E_d)]$$

A donor is ionised when its level is empty

$$N_d^+ \approx N_d e^{-(E_F - E_d)/kT}$$

Boltzmann form of $1 - f$*, since* $E_F - E_d \gg kT$

E_F between E_d and E_c



Check. Is E_F above or below E_d here, and what does that give?

- ✓ above, so $1 - f(E_d) \approx e^{-(E_F - E_d)/kT}$
- below, so $1 - f \approx 1$
- equal, so $1 - f = \frac{1}{2}$
- it does not matter

At moderate T, E_F lies between E_d and E_c, a few kT above E_d. Then $1 - f(E_d) = 1/(e^{(E_F - E_d)/kT} + 1) \approx e^{-(E_F - E_d)/kT}$.

EQUATE WITH N

$$N_c e^{-(E_c - E_F)/kT} = N_d e^{-(E_F - E_d)/kT}$$

substitute: $n = N_c e^{-(E_c - E_F)/kT}$

$$e^{(2E_F - E_c - E_d)/kT} = \frac{N_d}{N_c}$$

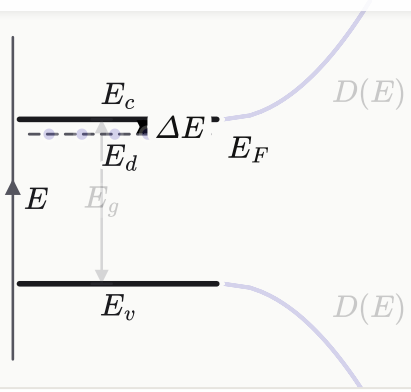
rearrange

$$E_F = \frac{E_c + E_d}{2} + \frac{kT}{2}\ln\frac{N_d}{N_c}$$

Take $\ln$ *and solve for* E_F

At moderate T, E_F lies about midway between E_d and E_c. As T rises ($N_c \propto T^{3/2}$) it moves down and finally reaches mid-gap: the sample turns intrinsic. Not valid at $T = 0$ or very high T.

$E_F \approx (E_c + E_d)/2$



ELECTRON CONCENTRATION

$$n = N_c \exp\left[-\frac{E_c - \frac{E_c+E_d}{2} - \frac{kT}{2}\ln\frac{N_d}{N_c}}{kT}\right]$$

substitute: E_F

$$n = N_c\left(\frac{N_d}{N_c}\right)^{1/2} e^{-(E_c-E_d)/2kT}$$

simplify

$$n = (N_d N_c)^{1/2} e^{-\Delta E/2kT}$$

simplify

$n \propto \sqrt{N_d}$ at moderate temperatures. Conductivity $\sigma = ne\mu_n \approx N_d e\mu_n$ once all donors are ionised. p-type, the same way: $E_F = \frac{E_v+E_a}{2} - \frac{kT}{2}\ln\frac{N_a}{N_v}$, $\sigma_p \approx N_a e\mu_p$.

$\Delta E = E_c - E_d$

